Heterogeneous Data-Based Global-to-Local Cross-Domain Few-Shot Classification Method of Hyperspectral Image

Shaoguang Huang, *Member, IEEE*, Kun Zhao, *Graduate Student Member, IEEE*, Hongyu Chen, *Member, IEEE*, and Hongyan Zhang, *Senior Member, IEEE*

***Abstract*—Cross-domain few-shot learning (FSL) has shown promising performance in hyperspectral image classification (HSIC) under limited labeled data. However, existing approaches often suffer from insufficient meta-knowledge transfer due to reliance on a single source domain, and fail to fully bridge the domain gap owing to the use of single-level alignment strategies. In this article, we propose a heterogeneous data-driven, global-to-local cross-domain FSL (CDFSL) framework for HSIC, leveraging richly labeled natural RGB images and hyperspectral data as source domains to support classification in the target domain with only a few labeled samples. The proposed method consists of two stages. In the first stage, we employ natural RGB images to learn a powerful spatial feature extractor with FSL and self-supervised learning, which will be fine-tuned for HSI in the second stage. To alleviate the domain gap between the source and target HSI domains at the second CDFSL stage, we propose a global-to-local domain adaptation (DA) strategy that performs alignment both at the domain level and class level, effectively reducing the learning bias toward the source HSI while promoting discriminative feature learning. In addition, to address large intraclass variance in the target HSI domain, we introduce a self-supervised contrastive loss based on positive pairs only, enhancing the within-class representation compactness. Extensive experiments on three benchmark datasets demonstrate that our method outperforms the state-of-the-art.**

***Index Terms*—Classification, cross-domain, few-shot learning (FSL), hyperspectral images (HSIs), remote sensing.**

NOMENCLATURE

Notations	Definition
$D_{\text{sn/sh}}$	Natural image/HSI in the source domain.
$D_{t/\text{tl/tu}}$	All/labeled/unlabeled HSI in the target domain.
F_s^a	Spatial feature extraction backbone network.
F_θ	Spatial–spectral feature extraction network (SSFEN) with parameter θ.
$S_{\text{sn/sh}/t}$	Support set in the source natural image domain/HSI domain/target domain.
$Q_{\text{sn/sh}/t}$	Query set in the source natural image domain/HSI domain/target domain.
P_k	kth prototype calculated by the support set.
$C_{s/t}^k$	kth class centroid in the source/target domain.
L_{fsl}	Loss of FSL.
L_{sn}^r	Loss of rotation prediction.
L_{mim}	Domain-level alignment loss.
$L_{\text{cl}}^{\text{ter}}$	Class-level alignment loss.
$L_{\text{cl}}^{\text{tra}}$	Target domain consistency (TDC) loss.

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The authors are with the School of Computer Science, China University of Geosciences (Wuhan), Wuhan 430074, China (e-mail: huangshaoguang@cug.edu.cn; 1627838940@qq.com; chenhongyu@cug.edu.cn; zhanghongyan@cug.edu.cn).

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) are 3-D data acquired by specialized imaging sensors, containing rich spatial and spectral information [1], [2], [3], [4]. With hundreds of continuous, narrow spectral bands, HSIs enable precise object identification and are widely used in applications such as land mapping [5], environmental monitoring [6], geological exploration [7], and medical research [8]. HSI classification (HSIC), as one of the basic and critical tasks in remote sensing interpretation, has received increasing attention.

Traditional HSIC methods are typically designed by using classical machine learning algorithms such as support vector machines (SVMs) [9], random forest [10], and extreme learning machines [11]. While theoretically grounded, their dependence on hand-crafted features creates practical limitations that hinder real-world deployment. In recent years, the rise of deep learning has significantly advanced HSIC, primarily due to the ability of these methods to automatically extract higher level, more abstract features. A variety of network architectures have been applied to HSIC, including convolutional neural networks (CNNs) [12], transformer networks [13], and Mamba networks [14]. CNNs excel at local feature extraction through their convolutional layers, while transformer networks leverage the self-attention mechanism to capture global dependencies across different spectral bands; the recent Mamba architecture, with its unique state-space model and efficient computational power, has proven

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effective for HSIC. In addition, numerous hybrid architectures have been proposed that combine various network models [15], [16], [17]. While these deep learning-based HSIC methods have demonstrated satisfactory performance, they typically rely on many labeled samples. As labeling HSI is time-consuming and expensive, the limited training data often makes the deep learning-based methods prone to overfitting [18].

Representative techniques to address the shortage of labeled training data include transfer learning [19], generative models [20], self-supervised learning [21], unsupervised learning [22], data augmentation [23], and few-shot learning (FSL). Among them, FSL [24] raises increasing attention due to its powerful generalization ability. It is inspired by the human cognitive process of recognizing new categories with the core idea of learning to learn. Meta-learning has emerged as a prominent paradigm in FSL, typically comprising two stages: meta-training and meta-testing. In the meta-training stage, the original dataset is restructured into a sequence of independent tasks (episodes), facilitating the model's acquisition of transferable meta-knowledge through the principle of "learning to learn" [25]. This knowledge is subsequently utilized in the meta-testing stage to perform classification on previously unseen classes. In recent years, FSL has been applied in the classification of HSI [26], [27], [28]. Directly applying FSL in HSIC often leads to unsatisfactory performance due to the very few number of categories and limited training data. Early research works [26], [29] construct meta-tasks by combining well-annotated source domain HSI to compensate for the scarcity of labeled data in the target HSI, which enables the model to learn more diverse spectral knowledge and improve the generalization ability of the model. However, significant differences in imaging sensors, imaging conditions, and land cover types between the source and target HSI domains often lead to substantial domain shifts, which in turn degrade model performance.

To solve this problem, a new cross-domain FSL (CDFSL) task is introduced for the HSI classification [30]. Depending on the number of source domains utilized, existing CDFSL-based HSI classification methods can be broadly divided into two categories: i) single-source-based methods [31], [32], [33], [34], [35], [36]; and ii) multisource-based methods [37], [38]. The former typically employ HSI as source domain data, which is captured at different scenes from the target HSI domain [31], [32], [34], [35]. Compared with the aforementioned FSL-based methods [26], [27], [28], these single-source CDFSL methods take into account the domain shift problem and achieve improved classification performance. For instance, Li et al. [31] propose a conditional adversarial domain adaptation (DA) framework for feature alignment, and Zhang et al. [32] leverage graph convolution networks (GCNs) with maximum mean discrepancy (MMD) to enhance interdomain alignment. In addition, recent attempts have been made by leveraging the rich labeled RGB dataset of natural images as the source domain to enhance the spatial feature extraction of HSI in the target domain [33], [36]. Specifically, Wang et al. [33] pretrain FSL models on large-scale RGB datasets and fine-tune them on target HSIs. Zhang et al. [39] improve prototype stability using mutual information (MI) and further mitigate domain shifts with MMD-based adaptation. As single-source CDFSL methods rely solely on meta-knowledge learned from a single source domain, they may struggle to achieve accurate classification in the complex target HSI domain, especially when labeled samples in the target domain are extremely scarce. Recently, multisource-based CDFSL methods have been proposed to enhance the models' cross-domain generalization ability by using different types of data in the source domain. For instance, Wang et al. [37] leverage HSI and LiDAR data from the same scene as the source domain data to construct a more expressive prototype space. Li et al. [38] regard label text information as text modality and integrate it with HSI in both source and target domains, which improves the semantic representation ability of image prototypes. The multisource-based CDFSL methods often show improved classification performance compared with the single-source-based CDFSL.

Despite the performance improvements, the following issues on CDFSL remain to be solved.

1) Most existing CDFSL-based classification methods for HSI rely on a single image modality as the source domain, which limits the learning of transferable meta-knowledge and hampers classification performance in the target HSI domain. Although recent studies [37], [38] have attempted to incorporate additional modalities such as LiDAR or label text as auxiliary source domain data, the limited availability of these modalities still constrains the effectiveness of meta-knowledge transfer.
2) Existing CDFSL-based classification methods for HSI typically alleviate the domain shift problem only at the domain level, which ignores the class-level shift across domains that might hinder useful meta-knowledge transfer to the target domain, leading to indiscriminative feature learning.
3) Under the scenario of extremely scarce labeled data, the intraclass differences within the target HSI can become more pronounced in the feature space due to the large within-class spectral variabilities, leading to fuzzy decision boundaries. This significantly impacts the performance of the prototype-based classification methods.

To this end, we propose a novel heterogeneous-data-based global-to-local CDFSL (HGL-CDFSL) framework for HSI. Given that natural RGB images with rich annotations are more readily available than remote sensing data, we leverage both natural RGB images and HSI in the source domain to facilitate the learning of generalizable meta-knowledge. The HGL-CDFSL consists of two stages. Specifically, in the first stage, we employ the rich labeled natural images to train a backbone for spatial feature extraction with self-supervised learning and FSL, which will be transferred and fine-tuned in the second CDFSL stage. In the second stage, we utilize labeled hyperspectral data from the source and target domains to train a model to yield domain-adaptive spatial–spectral representations for classification. Particularly, we propose a global-to-local DA method to alleviate the domain and class shift between the source and target HSI domains. At the global

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level, we develop entropy-based intradomain and interdomain losses, which promote discriminative feature extraction and cross-domain representation alignment. At the local level, we propose an interdomain class-level centroid matching (IDCM) strategy based on contrastive learning, which mitigates the negative meta-knowledge transfer caused by category disparity across domains, improving the model's generalization across domains. Furthermore, to address the large within-class spectral variabilities in the target HSI domain, we introduce a self-supervised contrastive learning method with only positive pairs to enhance the compactness of the class-specific representations, improving the robustness of the model, especially when only a few labeled data are available in the target HSI domain. Our experimental results on benchmark datasets demonstrate that our method outperforms the state-of-the-art.

Our main contributions can be summarized as follows.

1) We propose a two-stage CDFSL framework for HSIC that, for the first time, leverages both natural RGB images and HSIs as heterogeneous source domain data to support the classification in the target HSI domain. Compared with existing CDFSL approaches using heterogeneous sources, our method benefits from the broader availability and diversity of RGB images, while also effectively incorporating domain-specific spectral information from HSIs.
2) We propose a global-to-local DA strategy in the second CDFSL stage to enhance the discriminability of learned representations in the target HSI domain. This strategy addresses both domain-level shifts and class-level misalignment between the source and target HSIs, thereby facilitating more effective transfer of useful meta-knowledge and improving generalization in few-shot scenarios.
3) To address the significant spectral variability in the target HSI domain, we introduce a self-supervised learning approach that leverages only positive pairs for contrastive learning to promote the intraclass consistency. This design effectively stabilizes prototype-based classification, particularly under the constraint of limited labeled data in the target domain.

The remainder of this article is organized as follows. Section II briefly introduces the related works. Section III elaborates on our proposed method in detail. Section IV conducts extensive experiments and analyzes the experimental results. Finally, Section V concludes our work.

II. Related Work

A. CDFSL for HSIC

Traditional FSL tasks usually assume that all samples come from the same domain or distribution, but this assumption is often too idealistic in practical applications. In recent years, CDFSL [40] has attracted growing attention for its ability to effectively leverage labeled source domain data to support tasks in target domains with scarce labeled samples. Liu et al. [29] were the first to apply CDFSL methods to HSIC, utilizing a deep residual 3-D convolution network for feature extraction and integrating a prototype network to learn a metric space for classification. However, the work in [29] does not account for interdomain differences. Li et al. [31] were the first to propose a framework specifically tailored for CDFSL, which simultaneously learns embedded features from both source and target domains while employing a conditional adversarial DA strategy to jointly address FSL and domain shift. Building on this foundation, Hu et al. [41] proposed a lightweight feature extraction network to obtain more discriminative representations, while also optimizing category prototype computation and employing MMD to alleviate domain shift. More recently, researchers have recognized that disparities in label distributions across domains can introduce bias into model learning. To address this challenge, Qin et al. [30] applied disentanglement techniques to separate domain-specific and domain-invariant features, thereby leveraging shared representations to mitigate source domain bias. Similarly, Feng et al. [42] employed knowledge distillation alongside decoupled interaction modules in the distillation process to extract more task-relevant knowledge, thereby reducing the model's bias toward the source domain.

Recent studies have explored the use of heterogeneous data as source domain data to enhance CDFSL. For instance, Wang et al. [37] leveraged co-registered HSI and LiDAR data from the same scene to derive more fine-grained and stable prototypes, while Li et al. [38] employed textual label descriptions in HSI to align image and text modalities, thereby improving prototype representation. Although these approaches achieve higher classification accuracy compared to traditional single-source CDFSL methods, their real-world applicability remains constrained by the limited availability of source domain data, a consequence of high acquisition costs. Furthermore, existing methods inadequately address the critical issue of domain shift. For example, while Wang et al. [37] mitigate domain-level discrepancies through adversarial learning, they overlook class-level shifts across domains, and the adversarial training itself introduces optimization challenges. Similarly, Li et al. [38] do not explicitly account for domain shift in their framework. In addition, the performance of these methods is further hindered by large intraclass spectral variations, particularly in complex target domains with scarce labeled data. These limitations motivate our work, which proposes a novel global-to-local heterogeneous data-based CDFSL framework for HSI classification, designed to better address DA and intraclass variability challenges.

B. Domain Adaptation

DA [27] aims to address the distribution shift between the source and target domains, which can be used to reduce the performance drop when a model trained on one domain is applied to a different target domain. DA usually assumes that the source domain and the target domain are related [43], and can be divided into divergence-based, adversarial-based, and reconstruction-based methods. Divergence-based approaches are among the most widely adopted strategies. These methods aim to minimize domain discrepancy by aligning the statistical properties of feature distributions across domains, including the mean (first-order moment), covariance (second-order moment), and even higher order moments,

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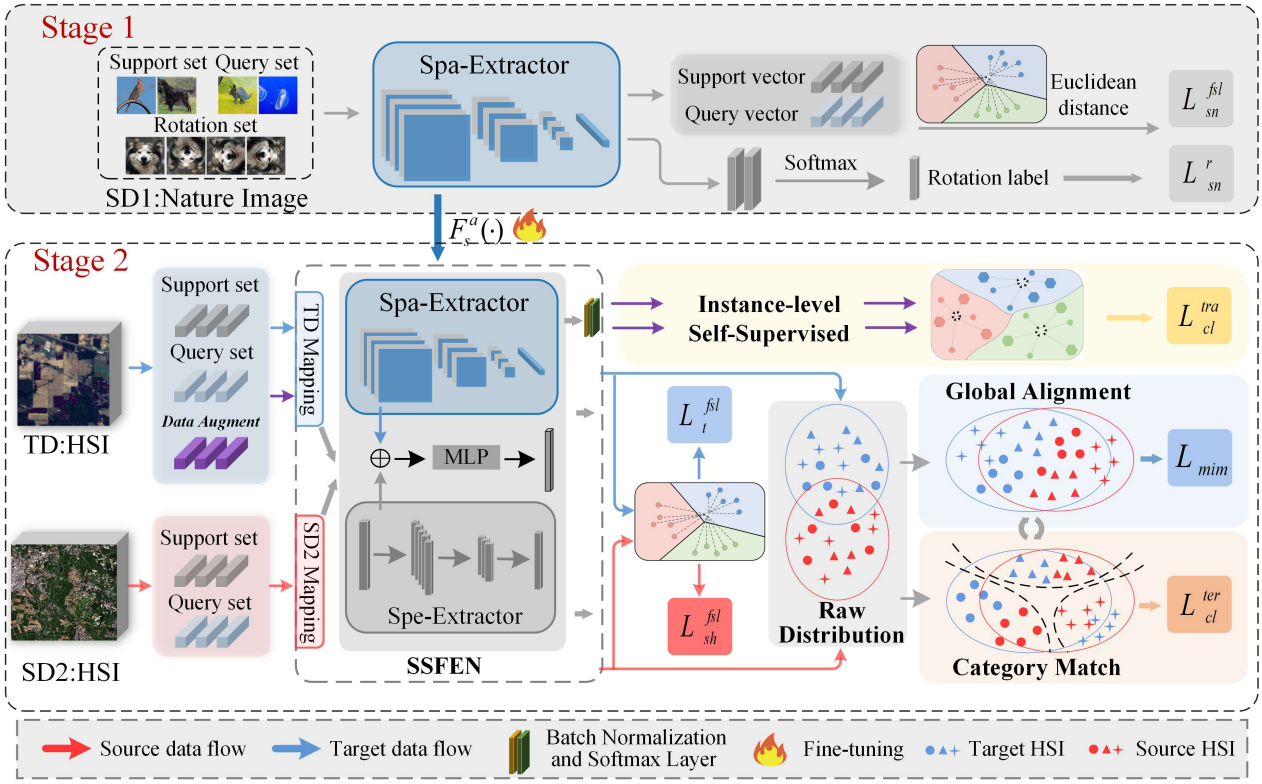


Fig. 1. Overview of the HGL-CDFSL. In the first stage, a spatial feature extractor is pretrained on natural images. In the second stage, CDFSL is performed on HSIs, which comprises three key components: 1) an MI-based strategy aims to mitigate network learning bias at the global level; 2) an IDCM strategy that ensures fine-grained alignment at the class level; and 3) instance-level contrast learning in the target domain, which enhances the consistent representation of samples within the same class.

which encourage the model to learn domain-invariant representations. Among these, the MMD method [44] aligns the mean feature distributions of the source and target domains in a reproducing kernel Hilbert space, and has inspired numerous extensions [45], [46]. Adversarial-based approaches employ adversarial training, where the feature extractor and domain discriminator are trained in opposition. For example, the domain adversarial neural network (DANN) [47] encourages domain-invariant representation learning by challenging the domain discriminator to distinguish between source and target features. Reconstruction-based approaches [48] facilitate DA by requiring the model to reconstruct input data, thereby promoting the learning of robust, domain-invariant features and implicitly aligning the feature spaces between the source and target domains. Recently, MI-based approaches [49], [50] have attracted increasing attention. These methods quantify and maximize the MI shared between domains, thereby enhancing feature correlation across domains and improving adaptation performance.

III. Proposed Method

Our CDFSL-based classification task of HSI is defined as follows. Let D_{sn} and D_{sh} denote the natural image source domain and the HSI source domain, respectively, and D_t represents the target HSI domain, where only a few labeled data points are available. D_t is further divided into two subsets: D_{tl}, which consists of labeled samples for training, and D_{tu}, which contains unlabeled samples for testing. We assume that the number of labeled data in the source RGB image and HSI domains is much larger than that in the target HSI domain, and the source and target domains have different classes. Our goal is to leverage the rich annotations in the source domains D_{sn} and D_{sh} to support the classification in the target domain D_t. The overall framework of our HGL-CDFSL is shown in Fig. 1, which mainly consists of two learning stages. The key notations are shown in Nomenclature.

In the first stage, we pretrain a spatial feature extraction backbone $F_s^a(\cdot)$ on D_{sn} with FSL and self-supervised learning. In the second stage, we leverage the labeled data from D_{sh} and D_{tl} to train a neural network for the classification of HSI in the target domain with FSL, where a shared SSFEN is designed to project the source and target HSIs into a common feature space. To alleviate the domain shift problem, we propose global and local representation alignment losses L_{mim} and $L_{\text{cl}}^{\text{ter}}$ for the feature alignment at both the domain and class levels, which facilitates discriminative feature learning. In addition, a view-consistency-based loss $L_{\text{cl}}^{\text{tra}}$ is introduced to deal with the large spectral variability problem in the target domain. A prototype-based method is used to obtain the classification results.

A. Spatial Feature Extraction Based on Fsl

Spatial information plays a crucial role in HSIC. Since natural images are more readily available with high-quality

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annotations and inherently contain richer spatial information than HSIs, we first pretrain a backbone neural network on natural images to obtain a strong spatial feature extractor. This extractor is then fine-tuned in the second stage to effectively capture the spatial features specific to HSIs. To acquire transferable spatial meta-knowledge from D_{sn}, FSL is used. During each meta-training task, we randomly select N classes from D_{sn}, and from each class, we randomly sample K labeled instances to form the support set $S_{\text{sn}} = \{(x_i, y_i)\}_{i=1}^{N\times K}$. In addition, we sample T labeled instances from the remaining data of the selected classes in D_{sn} to construct the query set $Q_{\text{sn}} = \{(x_j, y_j)\}_{j=1}^{N\times T}$.

We train the spatial feature extraction backbone $F_s^a(\cdot)$ using a metric-based meta-learning approach [51]. The probability of the sample x_j belonging to class k is obtained by

$$P\left(y_j = k \mid x_j \in Q_{\text{sn}}\right) = \frac{\exp\left(-d\left(F_s^a\left(x_j\right), P_k\right)\right)}{\sum_{k=1}^{N}\exp\left(-d\left(F_s^a\left(x_j\right), P_k\right)\right)} \tag{1}$$

where P_k is the prototype of the kth class in the support set S_{sn} and $d(\cdot)$ represents the Euclidean distance between two vectors. The cross-entropy-based FSL loss is formulated by

$$L_{s_n}^{\text{fsl}} = E_{S_{\text{sn}},Q_{\text{sn}}}\left[-\sum_{(x_j,y_j)\in Q_{\text{sn}}}\log P\left(y_j = k \mid x_j\right)\right]. \tag{2}$$

It is observed that the objects in remote sensing often have diverse orientations, while the orientation of objects in natural images is relatively consistent. This discrepancy limits the effective transfer of spatial meta-knowledge from D_{sn} to the spatial feature extraction process in HSIC. To enhance the robustness of the spatial feature extractor $F_s^a(\cdot)$ against different orientations of the objects in HSIs, we introduce a rotation-prediction auxiliary task [36]. Specifically, we rotate the images in D_{sn} with four directions $R = \{0°, 90°, 180°, 270°\}$, resulting in an expanded dataset D_{sn}^e. Let $x^r (r \in R)$ represent the rotated image of x in D_{sn}, and we use a prediction head R_φ, consisting of a fully connected layer and a softmax function, to predict the rotation category. The resulting cross-entropy loss is

$$L_{\text{sn}}^r = E_{x^r\sim D_{\text{sn}}^e}\left[\sum_{\forall r\in R} -\log R_\varphi^r\left(F_s^a\left(x^r\right)\right)\right] \tag{3}$$

where R_φ^r is the prediction score for the rotation r. The final loss at the first stage is

$$L_1 = L_{\text{sn}}^{\text{fsl}} + L_{\text{sn}}^r. \tag{4}$$

B. Source and Target HSI CDFSL

In the second stage, to leverage the spectral meta-knowledge from the richly labeled HSI in the source domain, we propose a CDFSL-based method to support HSIC in the target domain. The designed neural network is shown at the bottom of Fig. 1, which consists of a mapping layer and an SSFEN module. SD2 Mapping and TD Mapping are linear projection layers to ensure that the data from the two HSI domains have the same dimensions before being input into the SSFEN module, and their weights are not shared. The SSFEN module contains a spectral feature extractor and a spatial feature extractor, which are responsible for capturing spectral and spatial features of HSI, respectively. The outputs of the two branches are concatenated and passed through an MLP layer to obtain the final spatial–spectral feature. The input to the network at the second stage is a square patch centered at each pixel of an HSI. To capture spectral information, the spectral branch applies a 1D-CNN to the center pixel's spectral vector. For spatial feature extraction, the pretrained $F_s^a(\cdot)$ from the first stage is fine-tuned with HSI data. To fit with the input dimension of $F_s^a(\cdot)$, we employ a 2D-CNN to reduce the number of channels of the input 3-D HSI cube to three.

Similar to the first stage, we adopt a meta-learning training strategy and perform FSL on the source and target HSI domains in each episode, which aims to mine transferable meta-knowledge in D_{sh} and learn an embedding model suitable for D_t. By inputting the support set and the query set into the neural network denoted by $F_\theta(\cdot)$, the spatial–spectral embedding features are obtained. Taking D_{sh} as an example, with N-way K-shot, N categories are first randomly selected from D_{sh}, and K labeled samples are randomly selected from each category as the support set S_{sh}. Then, T labeled samples are randomly selected from the same category to construct the query set Q_{sh}. The category distribution of samples $x_{s,j}$ in the query set Q_{sh} is as follows:

$$P\left(y_{s,j} = k \mid x_{s,j} \in Q_{\text{sh}}\right) = \frac{\exp\left(-d\left(F_\theta\left(x_{s,j}\right), P_k^s\right)\right)}{\sum_{j=1}^{N}\exp\left(-d\left(F_\theta\left(x_{s,j}\right), P_k^s\right)\right)} \tag{5}$$

where P_k^s is the kth class prototype in S_{sh}, F_θ represents the spatial–spectral feature extractor with parameters θ, $x_{s,j}$ denotes the jth samples in Q_{sh} and $y_{s,j}$ is the prediction of $x_{s,j}$. Therefore, the loss function of the FSL on D_{sh} is defined as follows:

$$L_{\text{sh}}^{\text{fsl}} = E_{S_{\text{sh}},Q_{\text{sh}}}\left[-\sum_{(x_{s,j},y_{s,j})\in Q_{\text{sh}}}\log P\left(y_{s,j} = k \mid x_{s,j}\right)\right]. \tag{6}$$

Similar to (6), we define the FSL loss L_t^{fsl} in the target HSI domain D_t. The FSL training loss at the second stage is formulated as

$$L_{\text{fsl}} = L_{\text{sh}}^{\text{fsl}} + L_t^{\text{fsl}}. \tag{7}$$

C. Global-to-Local DA

The source and target HSI domains often show significant domain shift due to the disparity of land covers, imaging sensors, and imaging conditions. In addition, compared with the source domain, the number of labeled data in the target domain is very limited, which results in a learning bias problem due to the imbalanced supervision across domains. To address these problems, we develop a global-to-local DA loss, which facilitates discriminative feature learning while keeping domain and class alignment. To stabilize the model training at the early stage and mitigate the learning bias caused by domain imbalance, a warm-up strategy is applied before DA.

1) Domain-Level Feature Adaptation (DFA): We first propose a global DFA loss that maximizes intradomain MI while minimizing a domain alignment loss, thereby promoting the learning of discriminative, domain-invariant features.

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a) Maximize Intradomain MI: To enhance feature discriminability for classification, we maximize the MI between the input and its representations within both the source and target HSI domains, thereby encouraging more salient and sparse feature representations. Here, we take the target HSI domain as an example.

Let $z_t = F_\theta(x)$ be the extracted spatial–spectral features of $x \in D = S_t \cup Q_t$, where S_t and Q_t are the support set and query set in the target HSI domain, respectively. Then, we apply a softmax function to z_t and obtain the feature distribution z_{tgt}. The main idea here is to maintain the diversity of the representations within the domain while keeping the instance's representation sparse, which facilitates the learning of discriminative features for prototype-based classification. With the definition of MI [21], we maximize our objective function as follows:

$$\begin{aligned} I_{\text{tgt}} &= I\left(z_{\text{tgt}}; x\right) \\ &= H\left(p_{\text{tgt}}\right) - \mathbb{E}_{x\in D}\left[H\left(p\left(z_{\text{tgt}} \mid x;\theta\right)\right)\right] \end{aligned} \tag{8}$$

where $p_{\text{tgt}} = \mathbb{E}_{x\in D}[p(z_{\text{tgt}}|x;\theta)]$ and $H(\cdot)$ denotes the entropy function, defined as $H(p) = -\sum_i p_i \log(p_i)$, with p_i the ith entry of the distribution p. I_{tgt} is decomposed into two components: the entropy of the expected feature distribution of $F_\theta(\cdot)$, i.e., $H(\mathbb{E}_{x\in D}[p(z_{\text{tgt}}|x;\theta)])$, and the expected entropy of the feature distribution, i.e., $\mathbb{E}_{x\in D}[H(p(z_{\text{tgt}} \mid x;\theta))]$. Maximizing the first term promotes the diversity of representations in the target domain and minimizing the second term facilitates a sparse representation for each data point, thereby enhancing the discriminability of features. The proof of (8) is shown as follows:

$$\begin{aligned} &H\left(\mathbb{E}_{x\in D}\left[p\left(z_{\text{tgt}}|x;\theta\right)\right]\right) - \mathbb{E}_{x\in D}\left[H\left(p\left(z_{\text{tgt}} \mid x;\theta\right)\right)\right] \\ &= \mathbb{E}_{x\in D}\left[\sum_i p\left(z_{\text{tgt},i}|x\right)\log p\left(z_{\text{tgt},i}|x\right)\right] \\ &\quad - \sum_i \mathbb{E}_{x\in D}\left[p\left(z_{\text{tgt},i}|x\right)\log \mathbb{E}_{x\in D}\left[p\left(z_{\text{tgt},i}|x\right)\right]\right] \\ &= \mathbb{E}_{x\in D}\left[\sum_i p\left(z_{\text{tgt},i}|x\right)\log p\left(z_{\text{tgt},i}|x\right)\right] \\ &\quad - \mathbb{E}_{x\in D}\left[\sum_i p\left(z_{\text{tgt},i}|x\right)\log \mathbb{E}_{x\in D}\left[p\left(z_{\text{tgt},i}|x\right)\right]\right] \\ &= \mathbb{E}_{x\in D}\left[\sum_i p\left(z_{\text{tgt},i}|x\right)\log \frac{p\left(z_{\text{tgt},i}|x\right)}{\mathbb{E}_{x\in D}p\left(z_{\text{tgt},i}|x\right)}\right] \\ &= \mathbb{E}_{x\in D}\left[\int p\left(z_{\text{tgt}}|x\right)\log \frac{p\left(z_{\text{tgt}}|x\right)}{\mathbb{E}_{x\in D}p\left(z_{\text{tgt}}|x\right)}dz_{\text{tgt}}\right] \\ &= \int p(x)\,dx \int p\left(z_{\text{tgt}}|x\right)\log \frac{p\left(z_{\text{tgt}}|x\right)}{\int p(x)\,p\left(z_{\text{tgt}}|x\right)dx}dz_{\text{tgt}} \\ &= \int p(x)\,dx \int p\left(z_{\text{tgt}}|x\right)\log \frac{p\left(z_{\text{tgt}}|x\right)}{p\left(z_{\text{tgt}}\right)}dz_{\text{tgt}} \\ &= \int\int p\left(z_{\text{tgt}}, x\right)\log \frac{p\left(z_{\text{tgt}}, x\right)}{p\left(z_{\text{tgt}}\right)p(x)}dz_{\text{tgt}}dx = I\left(z_{\text{tgt}}; x\right). \end{aligned} \tag{9}$$

We summarize the calculation of I_{tgt} in Algorithm 1. The batch size in this algorithm corresponds to the size of $S_t \cup Q_t$ as specified in the experimental setup. Similar to I_{tgt} in the target domain, we maximize the following objective function in the source HSI domain:

$$I_{\text{src}} = H\left(p_{\text{src}}\right) - \mathbb{E}_{x\in D_{\text{sl}}}\left[H\left(p\left(z_{\text{src}} \mid x;\theta\right)\right)\right] \tag{10}$$

where $p_{\text{src}} = \mathbb{E}_{x\in D_{\text{sl}}}[p(z_{\text{src}}|x;\theta)]$, D_{sl} is the union of support and query sets in the source HSI domain and z_{src} is the feature distribution of $z_s = F_\theta(x)$ with $x \in D_{\text{sl}}$. Equation (10) works similar to (8) in the source HSI domain. The resulting sparse feature representations suppress irrelevant activations and enhance class separability by promoting confident and selective responses, thereby improving discriminability.

Algorithm 1 Pseudo-Code to Calculate the Target-Domain MI Loss in Each Episode

Input: target dataset D_t, episode number E_s, feature extractor $SSFEN = F_\theta(\cdot)$
for episode $e = 1$ to E_s **do**
  Sample an episode: $S_t, Q_t \subset D_t$
  Form minibatch: $D = S_t \cup Q_t$
  for each sample x in D **do**
    $z_t = F_\theta(x)$
    $z_{tgt} = \text{softmax}(z_t)$
  end for
  Compute $p_{tgt} = \mathbb{E}_{x\in D}[p(z_{tgt}|x;\theta)]$
  Compute $H(p_{tgt}) = -\sum_i p^i_{tgt}\log(p^i_{tgt})$
  Compute $I^e_{tgt} = H(p_{tgt}) - \mathbb{E}_{x\in D}[H(p(z_{tgt} \mid x;\theta))]$
end for
Output: Target-domain MI loss I^e_{tgt} for each episode

b) Domain-Level Representation Alignment: While maximizing intradomain MI improves the discriminability and diversity of features within each domain, the distribution gap between source and target domains still hinders effective knowledge transfer. To mitigate this issue, we propose a domain-level alignment strategy by minimizing the objective function as follows:

$$I_{\text{cross}} = H\left(p_{\text{all}}\right) - \frac{1}{2}\left(H\left(p_{\text{src}}\right) + H\left(p_{\text{tgt}}\right)\right) \tag{11}$$

where $p_{\text{all}} = p_{\text{src}} + p_{\text{tgt}}/(2)$ is the mean feature distribution across the source and target HSI domains. It is easy to check that when the mean feature distributions of the source and target domains are the same, i.e., $p_{\text{src}} = p_{\text{tgt}}$, (11) reaches its minimum. Thus, minimizing (11) promotes the mean feature representations of two domains to be closer, thereby achieving global domain-level representation alignment.

Combining (8), (10), and (11), we obtain the DFA loss as follows:

$$L_{\text{mim}} = -\left(I_{\text{src}} + I_{\text{tgt}}\right) + \lambda_{\text{cross}} \cdot I_{\text{cross}} \tag{12}$$

where λ_{cross} is a parameter to balance the two terms. We set $\lambda_{\text{cross}} = 0.5$ in our article.

2) Interdomain Class-Level Centroid Matching: In CDFSL, the source and target HSIs often exhibit significant differences in category sets and label spaces. Directly aligning global distributions across domains might lead to negative knowledge transfer, as it overlooks the category-specific variations and potentially introduces semantic mismatches. To mitigate this

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problem, we propose a class-aware alignment strategy that operates at a finer granularity, focusing on aligning feature distributions at the class level.

In each episode task, we first compute the centroid of each class from both the support and query sets in the source and target HSI domains in the representation space by $C_s^k = 1/(|\mathcal{N}_s^k|)\sum_{j\in\mathcal{N}_s^k} z_{s,j}$ and $C_t^k = 1/(|\mathcal{N}_t^k|)\sum_{j\in\mathcal{N}_t^k} z_{t,j}$, where $\mathcal{N}_s^k$ and $\mathcal{N}_t^k$ are the index sets of the kth class within support and query sets in the source and target domains, and $z_{s,j}$ and $z_{t,j}$ are the representations in the two domains, respectively. Then, we compute the similarity of class centroids C_s^k and C_t^j across domains by

$$D_{\text{kj}} = \frac{C_s^{k^T} C_t^j}{\|C_s^k\|_2 \|C_t^j\|_2}. \tag{13}$$

Let $\mathcal{C}_s$ and $\mathcal{C}_t$ represent the classes in the source and target HSI domains, and $\phi(\cdot)$: $\mathcal{C}_t \to \mathcal{C}_s$ represent a mapping function, which picks for each target class j the most similar source class. The mapping function is formulated by

$$\phi(j) = \arg\max_k \; D_{kj}. \tag{14}$$

Note that with (14) there might exist multiple mapping classes in the source domain for the jth class in the target domain. To ensure a unique one-to-one mapping, we compute the cosine similarity between class centroids and assign the class with the highest similarity in the source domain to the jth target class, denoted as $\phi^*(j)$.

With the matching results, we introduce a contrastive loss function (15) to enforce alignment between semantically similar categories in the source and target domains. Specifically, category pairs with high similarity are treated as positive samples, while unmatched categories serve as negative samples. Our contrastive loss to align the representations at the class level is shown as follows:

$$L_{\text{cl}}^{\text{ter}} = \sum_{i=1}^{N} -\log \frac{h\left(C_s^{\phi^*(i)}, C_t^i\right)}{\sum_{j=1}^{N} h\left(C_s^j, C_t^i\right) + \sum_{j=1, j\neq i}^{N} h\left(C_t^j, C_t^i\right)} \tag{15}$$

where $h(u, v) = \exp((u^\top v)/(\tau\|u\|_2\|v\|_2))$ and τ is the temperature parameter. By optimizing $L_{\text{cl}}^{\text{ter}}$, we encourage matched categories to be closer in the feature space while pushing apart mismatched ones, thereby reducing domain discrepancies and enhancing the model's generalization ability in the target domain.

D. TDC Strategy Based on Self-Supervised Contrastive Learning

To further enhance representation stability in the target domain, we introduce a self-supervised contrastive consistency strategy. In the target HSI, intraclass variations caused by sensor noise, spectral mixing, and image quality degradation often lead to unstable feature representations [52], which hinder prototype-based classification under FSL settings. Motivated by the effectiveness of self-supervised learning in capturing intrinsic structure without reliance on labeled data, we enforce consistency among features of the same sample under different augmentations. This promotes intraclass compactness and helps the model learn more robust and stable prototypes, thereby improving target domain classification performance.

Specifically, we apply two randomly selected data augmentation strategies to each sample in the target domain D_{tl} within each episode, generating two distinct augmented views. These views are then passed through the meta-learner $F_\theta(\cdot)$ to extract corresponding spatial–spectral features z_1 and z_2. After applying batch normalization and the softmax function, we obtain outputs p_1 and p_2, which serve as the basis for enforcing cross-view consistency. Our contrastive loss function is formulated as follows:

$$L_{\text{cl}}^{\text{tra}} = \frac{1}{2}D(p_1, z_2) + \frac{1}{2}D(p_2, z_1) \tag{16}$$

where $D(\cdot,\cdot)$ represents the negative of the inner product of two vectors.

Algorithm 2 TDC Calculation Pseudocode (PyTorch-Like)

Input: Minibatch data x from D_l, augmentation function aug, feature extractor $SSFEN$, function D
for Episode $e = 1$ to E_s **do**
  Apply augmentations: x_1, $x_2 = aug(x)$
  Concate batch: x_{aug} = torch.cat($[x_1, x_2]$, dim = 0)
  Obtain $z = SSFEN(x_{aug})$ and p = softmax(BN(z))
  Define batch size: $batch_size = x.shape[0]$
  Split augmented batch:
  z_1, $z_2 = z[: batch_size]$, $z[batch_size :]$
  p_1, $p_2 = p[: batch_size]$, $p[batch_size :]$
  Compute TDC loss:
  $L_{cl}^{tra} = \frac{1}{2}D(p_1, z_2) + \frac{1}{2}D(p_2, z_1)$
end for
Function D(p, z):
$z = z$.detach() ▸stop gradient
return $-(p * z).sum(\text{dim} = 1).mean()$
Output: TDC module loss L_{cl}^{tra}

To ensure stability and prevent representation collapse, motivated by [53], a stop-gradient (stopgrad) operation is adopted. Thus, (16) becomes

$$L_{\text{cl}}^{\text{tra}} = \frac{1}{2}D\left(p_1, \text{stopgrad}(z_2)\right) + \frac{1}{2}D\left(p_2, \text{stopgrad}(z_1)\right) \tag{17}$$

which indicates that z_2 and z_1 are constants in the first and second terms, respectively. The pseudo-code is shown in Algorithm 2.

Combining the FSL loss, the DA loss, and the target domain loss, the overall loss in the second stage is formulated as follows:

$$\begin{aligned} L_2 &= L_{\text{fsl}} + \left(\mu \cdot L_{\text{cl}}^{\text{ter}} + \eta \cdot L_{\text{mim}}\right) + L_{\text{cl}}^{\text{tra}} \\ &= L_{\text{fsl}} + L_{\text{cl}}^{\text{tra}} + \mu \cdot L_{\text{cl}}^{\text{ter}} \\ &\quad + \eta \cdot \left(-\left(I_{\text{src}} + I_{\text{tgt}}\right) + \lambda_{\text{cross}} I_{\text{cross}}\right) \end{aligned} \tag{18}$$

where μ and η are two parameters to control the contributions of the IDCM and TDC modules, and we set $\lambda_{\text{cross}} = 0.5$ in our article.

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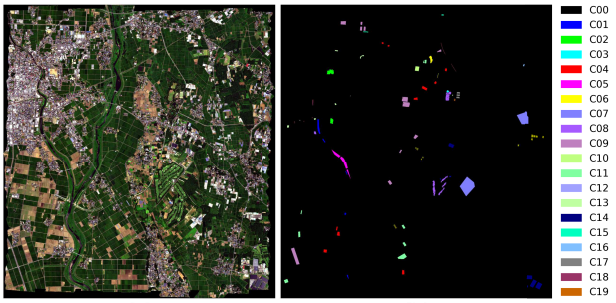


Fig. 2. Chikusei dataset. (a) False-color image. (b) Ground truth.

TABLE I
LAND COVER CLASS AND THE NUMBER OF SAMPLES FOR THE CHIKUSEI DATASET

Label	Color	Land Cover	Samples
C01		Water	2345
C02		Bare soil (school)	2859
C03		Bare soil (park)	236
C04		Bare soil (farmland)	48525
C05		Natural plants	4297
C06		Weeds in farmland	1108
C07		Forest	20516
C08		Grass	6515
C09		Rice filed (grown)	13369
C10		Rice filed (first stage)	1268
C11		Row crops	5961
C12		Plastic house	2193
C13		Manmade (non-dark)	1220
C14		Manmade (dark)	7664
C15		Manmade (blue)	431
C16		Manmade (red)	222
C17		Manmade grass	1040
C18		Asphalt	801
C19		Paved ground	145
		Total	77592

IV. EXPERIMENTS

A. Data Description

The datasets utilized in this article consist of source datasets with a sufficient number of labeled samples and target datasets with a limited amount of annotations. Building on the benchmarks established by previous studies, we select the Chikusei dataset [54] as the source HSI data and Mini-ImageNet [55] as the source natural RGB image data. For the target HSIs, we choose Indian Pines (IP) [56], Salinas (SA)[1], and WHU-Hi-HanChuan (HC) [57], [58]. This section provides a detailed description of each dataset.

1) Source Dataset: The MiniImageNet dataset is a subset of the ImageNet dataset, specifically used for FSL and meta-learning research. It contains 100 categories, 600 samples per category, and the size of each image is 84×84.

The Chikusei dataset was collected by a hyperspectral visible/near infrared camera equipped with the Headwall Hyperspec-VNIR-C sensor in Chikusei, Japan, in 2014. The Chikusei has a spatial size of 2517×2335 and with 19 labeled classes, as shown in Table I. The spatial resolution is 2.5 m per pixel, and it contains 128 spectral bands in the range of 363~1018 nm. Fig. 2 shows the false color image and the ground truth.

[1]www.ehu.eus/ccwintco/index.php/Hyperspectral Remote Sensing Scenes.

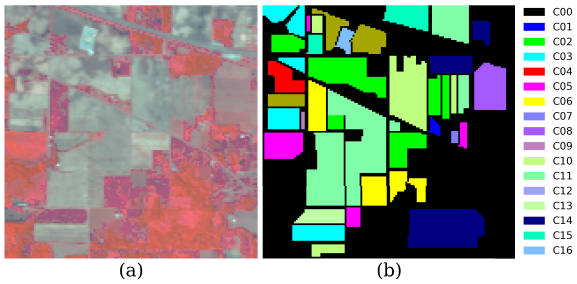


Fig. 3. Overview of the IP dataset. (a) False-color image and (b) ground truth.

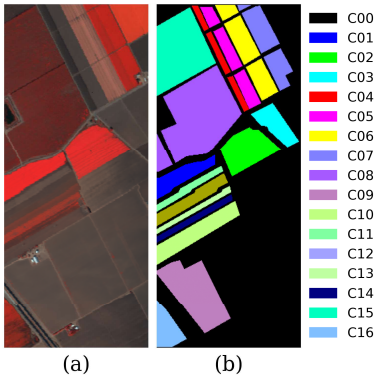


Fig. 4. Overview of the SA dataset. (a) False-color image and (b) ground truth.

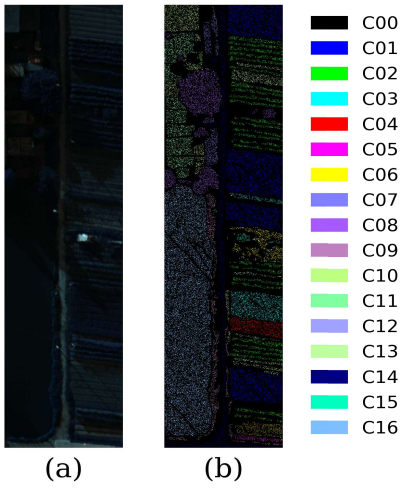


Fig. 5. Overview of the HC dataset. (a) False-color image and (b) ground truth.

2) Target Dataset: For the three target datasets, the annotated land cover types and the corresponding number of samples are listed in Table II. Specifically, the IP dataset was collected by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Indiana, USA, in 1992. It contains 200 spectral bands covering the 0.4–2.5-μm range, with a spatial resolution of 20 m/pixel and an image size of 145×145 pixels. Fig. 3 provides a visualization of the IP dataset. The

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TABLE II
NUMBER OF LABELED DATA IN DIFFERENT TARGET DATASETS

Label	Color	IP		SA		HC	
		Land Cover	Samples	Land Cover	Samples	Land Cover	Samples
C01		Alfalfa	46	Brocoli-green-weeds-1	2009	Stawberry	6710
C02		Corn-notill	1428	Brocoli-green-weeds-2	3726	Cowpea	3412
C03		Corn-mintill	830	Fallow	1976	Soybean	1543
C04		Corn	237	Fallow-rough-plow	1394	Sorghum	802
C05		Grass-pasture	483	Fallow-smooth	2678	Water Spinach	180
C06		Grass-trees	730	Stubble	3959	Watermelon	679
C07		Grass-pasture-mowed	28	Celery	3579	Greens	885
C08		Hay-windrowed	478	Grapes-untrained	11271	Trees	2696
C09		Oats	20	Soil-vinyard	6203	Grass	1420
C10		Soybean-notill	972	Corn-senesced-green-weeds	3278	Red roof	1577
C11		Soybean-mintill	2455	Lettuce-romaine-4wk	1068	Gray roof	2536
C12		Soybean-clean	593	Lettuce-romaine-5wk	1927	Plastic	551
C13		Wheat	205	Lettuce-romaine-6wk	916	Bare soil	1367
C14		Woods	1265	Lettuce-romaine-7wk	1070	Roaf	2784
C15		Buildings-Grass-Trees-Drivers	386	Vinyard-untrained	7268	Bright object	170
C16		Stone-Steel-Towers	93	Vinyard-vertical-trellis	1807	Water	11310
		Total	10249		54129		38622

SA dataset was also acquired using the AVIRIS sensor and comprises 204 spectral bands. It has a spatial resolution of 3.7 m/pixel and an image size of 512×217 pixels. Fig. 4 shows the visualization of the SA dataset. The HC dataset was captured in Hanchuan, Hubei, China, in 2016 using a headwall nano-hyperspec imaging sensor (17-mm focal length) mounted on a Leica Aibot X6 UAV V1 platform. The dataset consists of 274 spectral bands spanning the 0.4–1.0-μm range and has an image size of 1217×303 pixels. Due to the low solar elevation angle during the afternoon acquisition, the HC dataset includes extensive shadow-covered areas. A visualization of the HC dataset is shown in Fig. 5.

B. Experimental Setting and Evaluation Metrics

1) Implementation Details: In the first stage, we perform episode training on the entire Mini-ImageNet training set. The input patches for the natural images have a size of $33 \times 33 \times 3$. For each training episode, we construct an N-way, K-shot, T-query meta task, where N denotes the number of categories in the support set, and K denotes the number of samples per class within the support set. To simulate the FSL scenario, the number of support samples is set to $K = 1$. In addition, $T = 19$ query samples are randomly selected from the remaining images of the same N categories. We set N to 16 to match the number of categories in the target HSI domain. VGG16 is adopted as the backbone $F_s^a(\cdot)$ for spatial feature extraction, which aligns with the existing CDFSL method FSCF-SSL that uses natural images as the source domain. This choice not only ensures a consistent and fair experimental setup but also provides a simple and effective architecture.

In the second stage, 80 labeled samples are randomly selected from each class in the original Chikusei dataset to generate the meta trainset D_{sh}, enabling the learning of transferable spectral knowledge. Simultaneously, five labeled samples are randomly selected from each class in the target HSI dataset for training, and the remaining samples are used for testing. The meta-training and FSL processes are the same as in the first stage. The SD2 and TD mapping layers align the source and target domain HSI data to a unified dimensionality of 100. These layers act as necessary input adaptation modules rather than trainable components for performance improvement. Following the previous settings, we augment the target data by adding random Gaussian noise to construct S_t and Q_t. The data augmentation strategies in the TDC module are as follows.

1) Random Gaussian Noise: The original data are randomly scaled by a factor $\alpha \in (0.8, 1.2)$ and Gaussian noise with a standard deviation of 0.04 is added.
2) Random Flipping: The data are randomly flipped horizontally and vertically, each with a probability of 0.5.
3) Random Band Erasing: 10% of the spectral bands are randomly selected and set to zero to enhance spectral robustness.
4) Random Band Swapping: 10% of the spectral bands are randomly selected and swapped in random pairs to increase spectral diversity.

At the second stage, no data preprocessing, including spectral normalization and dimension reduction, is used before feeding the HSI to the neural network. For both stages, we use Adam as the optimizer, with a learning rate of 0.001. During pretraining on natural images, the number of training episodes is set to 900 for the IP dataset, 1100 for the SA dataset, and 1000 for the HC dataset. In the second stage, the number of training episodes is set to 1000 for all datasets. For the IP, SA, and HC datasets, we set μ and η for the IDCM and TDC modules to (0.8, 0.2), (0.2, 0.2), and (0.1, 0.1), respectively. To ensure stable training, we adopt a warm-up strategy for the IDCM and TDC modules. Their warm-up epochs are set to (0, 100), (150, 150), and (100, 100) in IP, SA, and HC, respectively. All experiments were conducted on a server equipped with an NVIDIA RTX 4090 GPU.

2) Compared Methods: To verify the performance of the proposed method, we compare our method HGL-CDFSL with eight SOTA methods on three benchmark datasets.

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TABLE III
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE IP DATASET

Label	SSRN [59]	DCFSL [31]	Gia-CFSL [60]	CMFSL [35]	FSCF-SSL [36]	FDFSL [30]	CTF-SSCL [61]	MLPA [62]	HGL-CDFSL
1	99.27±2.20	93.90±9.59	92.68±7.07	97.80±4.29	**99.51±0.98**	96.83±3.28	98.54±1.19	95.12±6.98	99.27±1.56
2	44.92±13.91	42.23±11.04	43.49±11.0	47.59±11.56	57.05±11.20	53.27±8.49	53.62±15.35	48.11±7.49	**63.64±9.23**
3	48.86±10.81	49.07±9.47	41.19±7.36	60.63±7.09	65.20±16.21	67.30±12.92	60.33±5.41	46.38±8.84	**70.91±11.60**
4	76.68±17.16	73.32±9.57	83.66±6.00	80.69±14.66	93.19±10.15	89.05±9.14	88.58±10.00	76.94±11.61	**94.66±5.86**
5	71.49±18.83	73.93±7.06	71.19±8.70	73.89±7.87	72.32±8.31	74.12±11.27	**79.08±8.62**	70.02±6.10	**79.08±7.85**
6	77.79±26.16	85.77±5.88	77.96±9.45	78.43±9.61	85.31±10.86	87.23±8.38	85.68±6.67	**87.37±4.75**	80.50±13.14
7	**100.00±0.00**	98.26±2.88	95.22±10.19	**100.00±0.00**	**100.00±0.00**	98.26±2.88	**100.00±0.00**	99.13±1.74	**100.00±0.00**
8	60.85±14.35	83.02±14.62	87.42±12.31	90.42±7.29	98.46±2.79	83.30±16.05	93.51±8.88	87.95±9.16	**99.30±1.82**
9	**100.00±0.00**	**100.00±0.00**	99.33±2.00	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.67±4.00	99.33±2.00	**100.00±0.00**
10	60.85±14.35	58.76±9.86	58.41±5.39	59.68±13.08	62.32±12.22	63.31±11.88	65.14±8.84	62.04±8.16	**68.16±12.34**
11	52.66±16.85	57.40±11.17	60.13±12.85	60.07±15.07	72.79±15.19	63.98±9.62	62.54±12.52	56.39±13.78	**79.27±8.87**
12	46.22±13.37	45.24±12.41	44.32±14.40	47.42±11.62	71.00±14.72	54.54±7.98	51.70±13.49	41.96±15.15	**73.52±15.21**
13	96.40±3.94	97.20±5.18	96.75±5.61	97.40±5.18	99.20±1.91	98.70±1.86	97.80±2.18	98.20±2.76	**99.80±0.33**
14	71.29±15.68	83.35±9.52	81.69±10.09	83.42±8.48	91.96±9.62	86.96±7.62	86.04±8.49	83.98±9.86	**94.79±5.42**
15	68.77±17.01	64.38±7.20	63.07±13.12	75.12±10.85	91.60±8.61	82.60±10.79	80.29±9.38	71.55±12.81	**95.77±7.96**
16	98.75±1.39	98.64±1.75	98.41±1.70	99.62±0.72	96.48±4.73	96.69±5.93	**99.77±0.45**	98.52±2.16	94.66±5.66
OA	60.87±5.00	63.30±3.25	62.89±2.73	66.28±1.76	75.42±4.04	70.39±2.37	70.07±2.76	64.40±3.24	**79.61±2.94**
AA	74.84±2.66	75.28±1.85	74.68±2.00	78.26±1.17	84.77±1.78	80.94±1.78	81.33±1.19	76.44±2.00	**87.08±1.64**
κ	56.18±5.21	58.75±3.52	58.18±2.81	61.95±2.88	72.19±4.41	66.73±2.51	66.31±2.91	59.97±3.37	**76.95±3.25**

Specifically, the spectral–spatial residual network (SSRN) [59], a popular deep learning-based supervised method, is used as a baseline to evaluate HSI classification performance in few-shot scenarios without meta-learning. The other seven methods are meta-learning-based approaches for HSI classification, including deep CDFSL (DCFSL) [31], graph information aggregation CDFSL (Gia-CFSL) [60], class-covariance metric FSL (CMFSL) [35], few-shot cross-domain learning with self-supervision for hyperspectral classification (FSCF-SSL) [36], feature disentanglement-based FSL (FDFSL) [30], semi-supervised contrastive CNN-Transformer framework (CTF-SSCL) [61], and multilevel prototype alignment FSL method (MLPA) [62]. To ensure a fair evaluation, FSCF-SSL uses natural images as the source domain data, and we keep our corresponding setting at the first stage the same as FSCF-SSL. In the six meta-learning-based HSI classification methods DCFSL, Gia-CFSL, FDFSL, CMFSL, MLPA, and CTF-SSCL, the Chikusei dataset serves as the source domain HSI.

3) Evaluation Metrics: To eliminate estimation bias, all experiments were independently conducted ten times. For performance evaluation, we use three commonly used metrics in HSIC tasks: overall classification accuracy (OA), average classification accuracy (AA), and the kappa coefficient (κ). The average of ten runs is reported.

C. Classification Performance

We show the results on different target HSI datasets in Tables III–V. The highest classification accuracy is marked in bold, and the second-best is underlined.

1) Results on the Ip Dataset: Table III shows the classification results of different methods on the IP dataset, and the corresponding classification maps are shown in Fig. 6. Overall, our method achieves superior performance across all metrics OA, AA, and κ, reaching 79.61%, 87.08%, and 76.95%, respectively, with lower standard deviations. The IP dataset exhibits a highly imbalanced class distribution. In scenarios with limited labeled samples, FSL-based methods consistently outperform the conventional deep learning approach SSRN. Among the FSL-based methods, DCFSL is the first to integrate FSL with adversarial adaptation to mitigate interdomain discrepancies. Compared to the robust deep learning method SSRN, DCFSL improves OA, AA, and Kappa by 2.43%, 0.44%, and 2.57%, respectively. Gia-CFSL employs graph information aggregation for relational modeling to achieve cross-domain alignment. However, the lack of explicit constraints on nonlocal feature propagation can lead to noisy relationship modeling, which limits its performance gains. However, these methods fail to fully exploit the spatial information inherent in HSI data. In contrast, FSCF-SSL leverages natural images as the source domain to enhance the model's ability to capture spatial features of HSI, resulting in improved performance across all metrics. Our method leverages both natural images and HSI as source domains, enabling the extraction of more transferable meta-knowledge. Compared to FSCF-SSL, which relies solely on natural images, our method further improves OA, AA, and Kappa by 4.19%, 2.31%, and 4.76%, respectively, and achieves the highest or second-highest accuracy in 14 categories. The visual comparisons in Fig. 6 further confirm that our method better aligns with the ground truth, especially in accurately classifying underrepresented classes (e.g., classes 7, 9, and 16) and visually similar categories such as different types of corn (classes 2–4).

2) Results on the SA Dataset: Table IV presents the classification results of each method on the SA dataset, with the corresponding visualizations shown in Fig. 7. In general, our method HGL-CDFSL achieves the best performance in terms of OA, AA, and κ, reaching 92.75%, 95.64%, and 91.94%, respectively. The SA dataset features a relatively balanced category distribution, and the traditional deep learning-based method SSRN yields an OA of 80.51%. Similar trends to those observed on the IP dataset can be found here. FSL-based methods outperform the conventional deep learning method SSRN, which demonstrates the advantage of

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TABLE IV
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE SA DATASET

Label	SSRN [59]	DCFSL [31]	Gia-CFSL [60]	CMFSL [35]	FSCF-SSL [36]	FDFSL [30]	CTF-SSCL [61]	MLPA [62]	HGL-CDFSL
1	89.06±29.77	98.78±1.35	99.07±0.87	98.51±0.94	99.63±0.75	98.51±2.09	**99.84±0.22**	98.81±2.19	98.70±3.76
2	60.92±36.58	99.40±0.90	98.74±1.77	99.03±1.16	97.02±3.25	99.04±2.41	98.91±1.10	99.01±2.04	**99.68±0.63**
3	84.12±11.54	92.40±10.29	90.36±9.96	94.00±5.76	**96.68±5.14**	93.30±10.81	93.94±5.81	90.60±9.89	92.21±10.35
4	99.55±0.69	99.30±0.54	99.30±0.63	99.42±0.61	**99.92±0.13**	99.14±0.74	99.06±1.52	99.22±0.90	99.14±2.40
5	94.57±4.63	91.43±4.50	90.10±6.38	91.85±3.65	**96.15±2.46**	91.81±3.21	93.24±4.07	91.84±3.18	95.73±3.25
6	98.50±1.18	99.00±0.79	98.87±1.06	99.22±0.67	99.58±0.62	99.14±0.89	99.54±0.41	99.27±0.59	**99.73±0.30**
7	97.46±7.09	98.74±2.64	98.71±1.83	99.07±0.87	97.71±6.15	98.85±1.10	99.38±0.90	98.35±1.62	**99.47±0.54**
8	62.45±22.66	74.82±12.71	73.11±12.74	**95.86±9.08**	78.67±3.94	80.17±4.36	76.02±8.89	76.45±9.77	83.29±5.42
9	91.21±13.23	98.84±1.13	98.70±2.69	99.34±1.11	97.80±2.64	98.79±1.01	99.28±1.58	98.34±2.00	**99.76±0.31**
10	74.13±15.11	83.11±9.32	80.18±13.34	85.20±8.06	**94.33±4.25**	83.31±9.24	81.02±8.87	83.88±7.81	91.27±5.74
11	94.12±6.29	98.31±2.36	96.33±3.44	97.81±2.77	**99.51±0.78**	96.98±2.75	97.51±1.78	98.01±2.05	98.16±2.87
12	96.06±5.01	**99.57±0.51**	99.03±1.04	97.58±6.66	99.13±1.20	98.94±1.68	98.99±1.17	98.87±0.96	98.81±1.75
13	97.24±4.11	99.22±0.43	97.83±1.31	**99.61±0.35**	99.51±0.38	97.78±1.81	98.75±1.55	98.60±1.86	99.23±0.65
14	97.35±1.76	98.51±1.45	97.92±1.65	**99.08±0.52**	97.93±1.94	98.79±1.26	95.72±7.02	98.73±0.87	96.14±3.48
15	69.23±23.22	69.40±10.25	75.93±6.59	77.14±8.15	78.20±12.08	82.24±8.33	76.96±8.31	72.72±6.82	**83.21±6.46**
16	88.82±6.27	90.29±5.06	87.09±8.38	85.94±7.76	**97.80±2.37**	92.64±4.85	93.41±5.94	89.73±4.87	95.71±6.56
OA	80.51±5.56	88.11±1.70	87.98±2.54	87.97±1.76	91.15±2.64	91.00±1.28	89.53±1.86	88.75±1.35	**92.75±1.04**
AA	87.18±3.96	93.19±0.91	92.54±0.71	93.09±1.13	95.60±1.62	94.34±1.12	93.85±1.06	93.28±1.05	**95.64±1.09**
κ	78.46±6.05	86.79±1.85	86.66±2.79	86.65±1.92	90.17±2.95	89.99±1.41	88.38±2.04	87.50±1.47	**91.94±1.15**

TABLE V
CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HC DATASET

Label	SSRN [59]	DCFSL [31]	Gia-CFSL [60]	CMFSL [35]	FSCF-SSL [36]	FDFSL [30]	CTF-SSCL [61]	MLPA [62]	HGL-CDFSL
1	61.38±20.85	71.10±8.87	71.42±8.93	77.06±8.67	75.03±10.11	76.35±9.22	**77.47±8.31**	76.61±8.11	74.94±6.05
2	28.27±15.21	46.03±8.23	41.17±8.68	58.88±8.93	**85.57±4.33**	44.25±11.10	51.44±8.66	49.77±7.73	79.34±11.07
3	34.27±22.31	51.66±9.67	49.49±15.27	72.44±12.50	67.74±18.06	62.09±10.65	76.80±8.46	47.85±13.41	**78.79±8.72**
4	65.17±33.29	90.80±5.48	84.97±11.18	95.11±2.28	80.16±20.38	88.12±9.35	**96.11±2.36**	88.71±7.64	82.12±10.00
5	75.60±23.25	86.97±5.29	82.63±9.04	89.09±8.20	**98.51±1.52**	89.31±7.94	96.97±3.59	86.40±8.16	**98.51±2.48**
6	24.61±13.45	25.80±11.01	29.97±13.53	38.18±8.15	44.53±12.33	30.01±9.73	44.87±5.08	24.38±8.27	**45.62±9.20**
7	58.74±26.73	**82.57±11.55**	74.61±5.77	80.61±12.06	77.17±12.13	75.59±8.83	81.15±7.53	77.95±10.77	67.78±15.00
8	31.15±12.63	33.74±7.43	37.02±10.27	47.80±12.43	46.91±14.46	43.80±9.60	51.81±9.64	32.68±8.23	**64.96±14.95**
9	34.97±10.99	30.00±10.69	31.85±9.95	43.36±10.80	47.46±14.13	38.54±14.57	**51.89±13.33**	31.72±9.63	49.18±13.99
10	52.59±16.51	71.61±9.91	67.65±14.91	82.42±10.61	54.25±15.21	74.29±7.54	**82.58±12.86**	67.05±12.69	53.00±18.43
11	49.29±29.12	71.85±9.31	63.40±15.65	70.73±8.77	51.16±15.96	57.64±9.27	**79.31±8.56**	68.64±7.46	62.05±20.61
12	44.96±17.56	48.41±10.22	46.10±11.22	53.55±9.50	75.88±14.35	55.79±15.16	55.35±15.59	49.60±13.56	**76.19±14.31**
13	30.77±15.91	42.28±7.09	36.28±10.73	42.33±6.16	45.84±10.06	33.77±5.43	49.30±8.08	38.39±10.39	**50.15±10.89**
14	34.92±16.78	64.26±10.39	62.04±12.34	61.49±12.38	**79.09±7.88**	59.76±12.27	60.55±12.55	62.54±11.84	73.83±9.83
15	63.39±21.81	69.03±10.03	68.91±11.09	78.73±8.43	**82.24±9.84**	63.45±11.26	81.27±5.52	70.97±11.24	79.39±11.56
16	78.89±25.85	87.62±10.05	88.01±11.65	84.44±11.47	**94.05±4.30**	90.46±9.48	88.30±10.98	86.53±11.20	92.66±4.17
OA	54.50±7.20	67.01±3.17	65.59±2.93	71.10±4.42	74.89±4.51	68.54±3.76	73.39±3.94	67.01±3.20	**75.96±3.13**
AA	48.06±4.78	60.86±1.93	58.47±3.05	67.24±2.05	69.10±3.44	61.45±3.22	70.32±2.63	59.99±1.40	**70.53±3.76**
κ	48.28±7.22	62.16±3.36	60.47±3.06	66.86±4.74	70.87±5.06	63.74±4.02	62.05±3.33	69.45±4.27	**72.18±3.57**

FSL in scenarios with limited labeled data. FSCF-SSL, which uses natural images as the source domain, outperforms the FSL-based methods DCFSL, Gia-CFSL, FDFSL, and CTF-SSCL, which use HSI as the source domain. This performance gain can be mainly attributed to the enhanced spatial feature extraction capability enabled by the meta-knowledge learned from natural images. Our method leverages both natural image and HSI as source domain data, leading to improved performance. Although FDFSL employs a low-rank orthogonal constraint for feature disentanglement, completely separating domain-common and domain-specific components in the high-dimensional HSI feature space remains challenging, which may limit its generalization ability on the target domain. Compared to FDFSL, our method improves OA, AA, and κ by 1.75%, 1.3%, and 1.95%, respectively; compared to FSCF-SSL, the improvements are 1.6%, 0.04%, and 1.77%, respectively. The SA dataset includes several spectrally similar classes, such as broccoli (classes 1–2) and lettuce (classes 11–14). Leveraging multilevel prototype alignment, MLPA obtains good performance on classes 4 and 6. Our method achieves the highest or second-highest classification accuracy in 9 out of the 16 classes. As shown in Fig. 7, the classification maps produced by our method are smoother, with fewer noisy predictions for most classes.

3) Results on the HC Dataset: Table V reports the classification results of all methods on the HC dataset. In general, all methods perform similarly with the results in Tables III and IV. With limited training data, SSRN yields an OA of 54.50% with a standard deviation of 7.20, which is not stable due to the effect of randomly selected training data. All other FSL-based methods show improved OA and standard deviation. Compared to DCFSL and Gia-CFSL, which rely on Euclidean distance metrics, CMFSL achieves better

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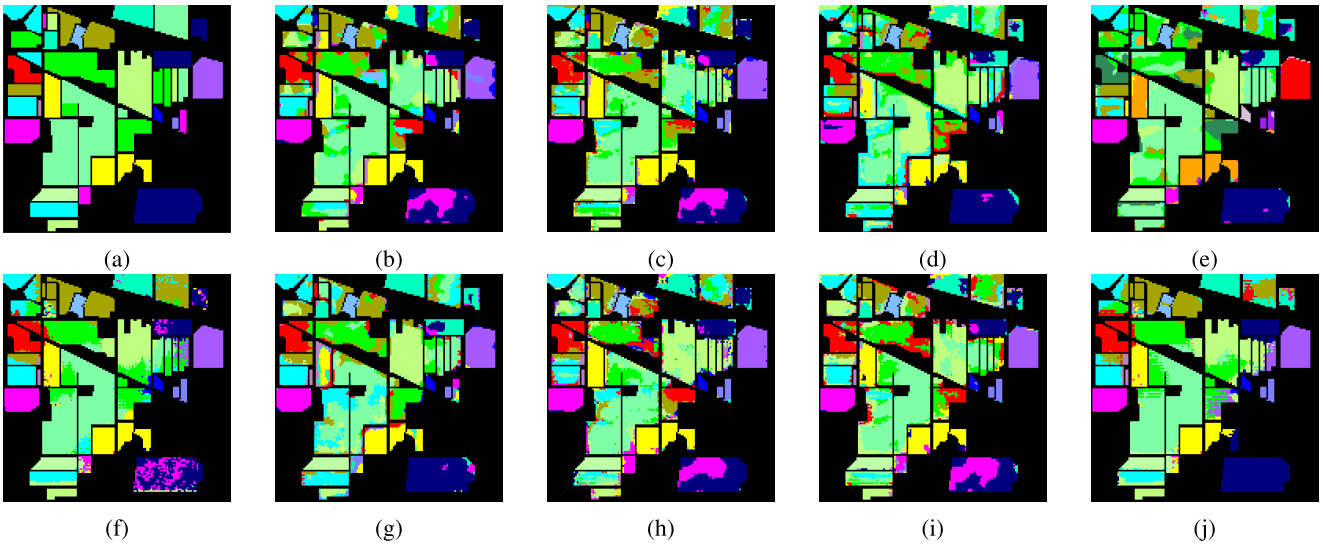


Fig. 6. Classification visualization maps for different methods on the IP dataset. (a) Ground-truth map, (b) SSRN, (c) DCFSL, (d) Gia-CFSL, (e) CMFSL, (f) FSCF-SSL, (g) FDFSL, (h) CTF-SSCL, (i) MLPA, and (j) HGL-CFSL.

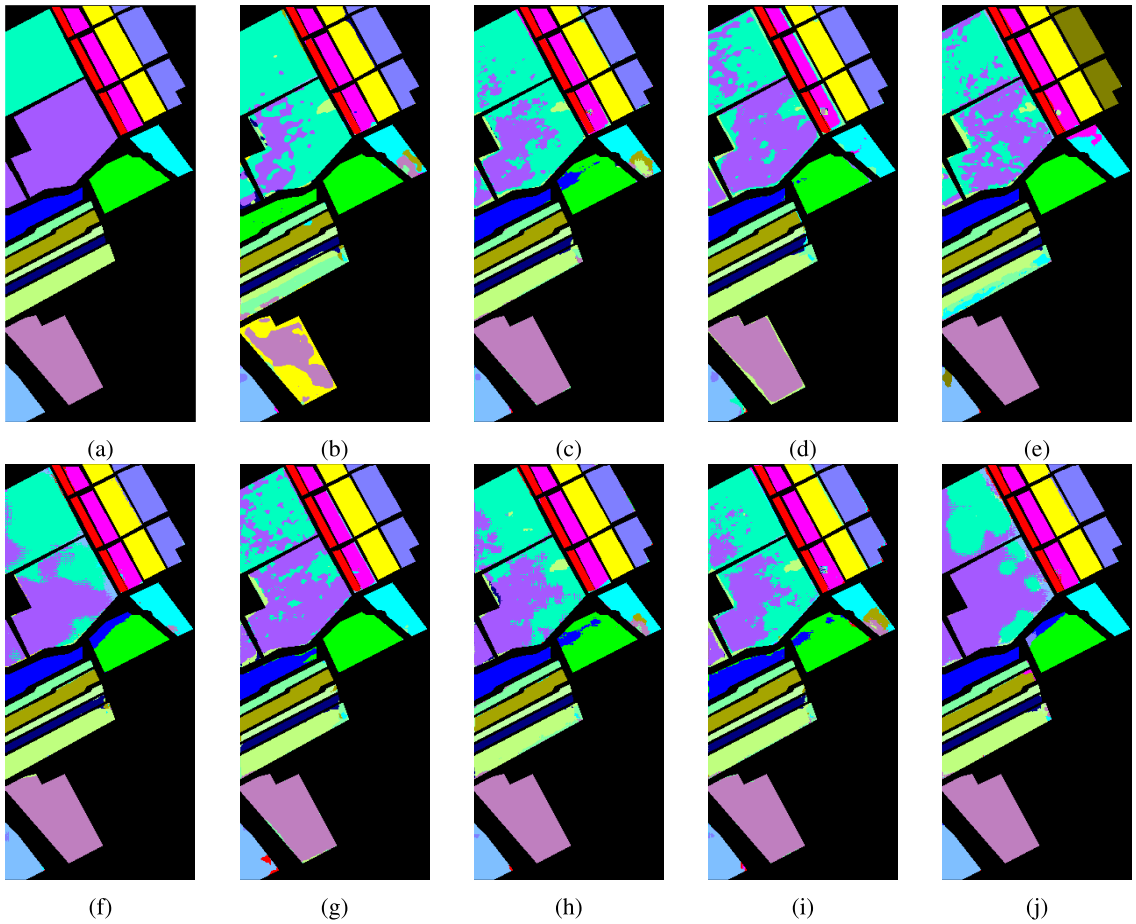


Fig. 7. Classification visualization maps for different methods on the SA dataset. (a) Ground-truth map, (b) SSRN, (c) DCFSL, (d) Gia-CFSL, (e) CMFSL, (f) FSCF-SSL, (g) FDFSL, (h) CTF-SSCL, (i) MLPA, and (j) HGL-CFSL.

performance by leveraging Mahalanobis distance. Compared with DCFSL, CTF-SSCL benefits from efficient feature extraction through the combination of CNN and Transformer architectures, resulting in OA, AA, and Kappa improvements of 6.38%, 9.46%, and 7.29%. However, it does not explicitly address interdomain distribution discrepancies, which limits its generalization capability when applied to cross-scene or cross-sensor HSI data, thereby constraining further

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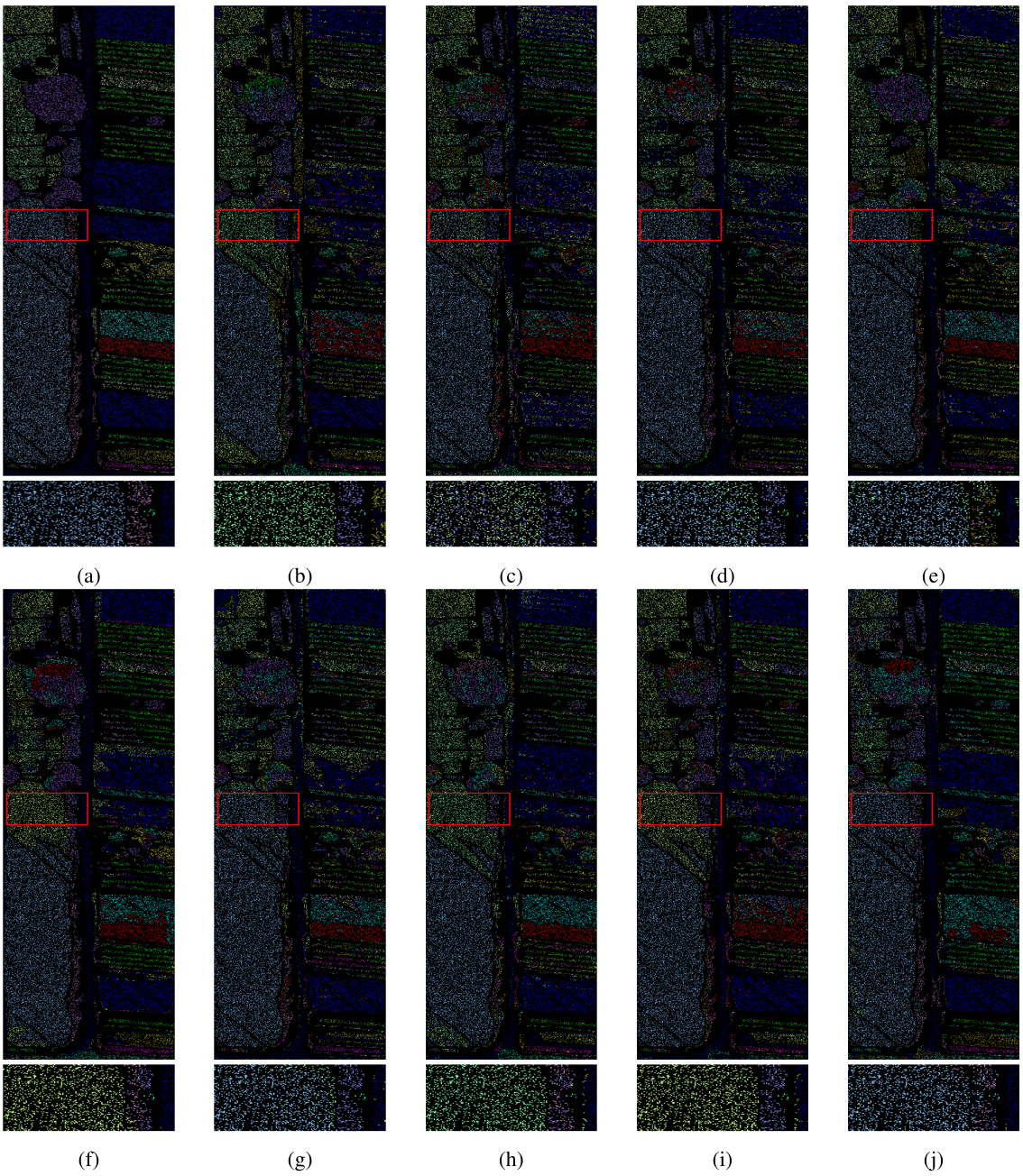


Fig. 8. Classification visualization maps for different methods on the HC dataset. (a) Ground-truth map, (b) SSRN, (c) DCFSL, (d) Gia-CFSL, (e) CMFSL, (f) FSCF-SSL, (g) FDFSL, (h) CTF-SSCL, (i) MLPA, and (j) HGL-CFSL.

performance improvement. Although MLPA alleviates DA issues via multilevel prototype alignment, its prototypes represent only class means and fail to adapt to the complex intraclass spectral variability in HSI, resulting in lower accuracy. Consistently, HGL-CDFSL and FSCF-SSL perform better than others due to the exploitation of richly labeled natural RGB images, leading to improved spatial feature extraction. Compared with the single-source-based FSL methods, our method incorporates heterogeneous data and thereby yields better results. The visual results in Fig. 8 indicate that the classification map generated by our method most closely aligns with the ground truth, particularly in the left region of the scene.

D. Ablation Study

To evaluate the influence of different components in HGL-CDFSL, we conduct ablation experiments here. The key components include the FSL strategy, the interdomain category matching (IDCM) module, the DFA module, and the TDC module. Accordingly, we design eight variants from Scheme 1 to Scheme 8 as shown in Table VI, where Scheme 8 refers to the proposed HGL-CDFSL.

We show the results on different datasets in Table VI. It is evident that the performance of our method improves progressively with the inclusion of each additional component. As expected, Scheme 8, combining all modules, achieves

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TABLE VI
ABLATION STUDY

Schemes	FSL	IDCM	DFA	TDC	IP			SA			HC		
					OA	κ	R	OA	κ	R	OA	κ	R
1	✓	✗	✗	✗	75.56±2.25	72.32±2.59	0.2453	89.49±2.23	88.33±2.46	0.1449	72.04±2.85	67.80±3.01	0.3933
2	✓	✓	✗	✗	75.69±2.30	72.51±2.62	0.2432	89.59±2.73	88.44±2.99	0.1409	72.18±3.47	67.96±3.68	0.3928
3	✓	✗	✓	✗	76.13±2.50	73.00±2.83	0.2437	90.24±2.36	89.13±2.64	0.1375	72.77±2.20	68.49±2.47	0.3954
4	✓	✗	✗	✓	78.76±3.42	75.95±3.81	0.2450	91.48±2.07	90.53±2.29	**0.1297**	75.10±3.51	71.25±3.89	0.3800
5	✓	✓	✓	✗	75.87±1.45	72.73±1.57	0.2353	89.67±2.25	88.53±2.48	0.1455	73.48±2.48	69.29±2.80	0.3975
6	✓	✗	✓	✓	78.05±4.28	75.15±4.83	0.2496	91.87±1.09	90.95±1.22	0.1306	73.93±3.66	69.97±3.98	**0.3707**
7	✓	✓	✗	✓	78.30±2.40	75.48±2.68	0.2434	91.91±1.24	91.01±1.38	0.1320	74.48±3.89	70.57±4.17	0.3657
8	✓	✓	✓	✓	**79.61±2.94**	**76.95±3.25**	**0.2351**	**92.75±1.04**	**91.94±1.15**	0.1321	**75.94±3.07**	**72.10±3.35**	0.3756

TABLE VII
CLASSIFICATION ACCURACY (OA) WITH THE FROZEN AND FINE-TUNED SPATIAL FEATURE EXTRACTORS

Datasets	Frozen model	Fine-tuned model
IP	39.27±3.17	79.61±2.94
SA	70.86±3.24	92.75±1.04
HC	45.46±2.73	75.94±3.07

the highest classification performance across all three target datasets. Notably, the DA strategy that integrates both the IDCM and DFA modules yields more performance gains than incorporating either module individually. This result demonstrates the effectiveness of our approach in addressing fine-grained DA, encompassing both global alignment and local refinement. Furthermore, the TDC module contributes the most significant improvement among all components, highlighting the critical role of intraclass consistency within the target domain. In summary, the combination of these modules and learning strategies significantly enhances the model's classification accuracy and generalization capability under FSL scenarios.

We also report the ratio of intraclass to interclass distances (R) in Table VI as a metric of class separability. Specifically, the intraclass distance is the mean Euclidean distance from each sample to its class centroid C_t^k, and the interclass distance is computed as the mean centroid-to-centroid distance across all class pairs. A smaller R value generally indicates greater class separability. It is observed that each module (i.e., IDCM, DFA, or TDC) generally reduces the value of R compared to the baseline FSL setting, indicating enhanced feature discriminability. However, it is noteworthy that the module combinations achieving the highest OA do not always correspond to the smallest R values. This discrepancy arises because while a smaller R generally reflects better feature-space separability, it does not solely determine classification performance. Factors such as decision boundary alignment, class imbalance, and model generalization capability also significantly influence the final OA.

To verify the effectiveness of the fine-tuning strategy of the pretrained spatial feature extractor, we conduct experiments on three datasets by freezing the spatial extractor of our method at the second stage. The results shown in Table VII indicate a significant performance drop when the spatial feature extractor is frozen, demonstrating that fine-tuning the pretrained model to adapt to HSI is highly important.

E. Parameter Sensitivity Analysis

We conduct experiments on three datasets to investigate the influence of the parameters μ and η on the performance of our method. In addition, we investigate the impact of warm-up epoch duration on model performance.

μ is tuned within the range [0.1, 1.0], while the warm-up phase spans 0–250 epochs. The results are shown in Fig. 9. It is observed that our method performs differently on different datasets. In the IP dataset, the performance of our method is more sensitive to the warm-up epoch than μ. In the SA dataset, our model performs quite stably for the two parameters. In the HC dataset, the performance of our method is stable at certain ranges for the two parameters. Based on the results, the optimal settings for μ and the warm-up epoch are (0.8, 0), (0.2, 200), and (0.1, 100) on the three datasets. It should be noted that the results in Fig. 9 with different parameters are better than the compared methods in Tables III–V in most cases, demonstrating the effectiveness of our method.

Similarly, we tune η in the range [0.1, 1.0] and the warm-up epoch within the range {0, 50, 100, 150}. The results are shown in Fig. 10. In the SA and HC datasets, our method is more sensitive to η than the warm-up epoch. In the IP dataset, the warm-up epoch often affects more the performance of our method than η. Based on the results, the optimal settings for η and the warm-up epoch are (0.2, 100), (0.2, 150), and (0.1, 100) on the three datasets. We also notice that even with suboptimal parameters, our method often outperforms the compared methods in Tables III–V.

To evaluate the robustness of our method, we conduct a comprehensive analysis of the hyperparameters μ and η by testing their different combinations. μ and η serve as balancing factors for the IDCM and DFA losses, where η explicitly weights the MI loss in (12). As illustrated in Fig. 11, our approach demonstrates consistent performance across a wide range of μ and η values. The optimal values of μ and η vary across datasets, mainly due to differences in land cover types, spatial resolutions, and class variability. Notably, the proposed method maintains stable accuracy improvements over the SOTA methods in Tables III–V in most parameter

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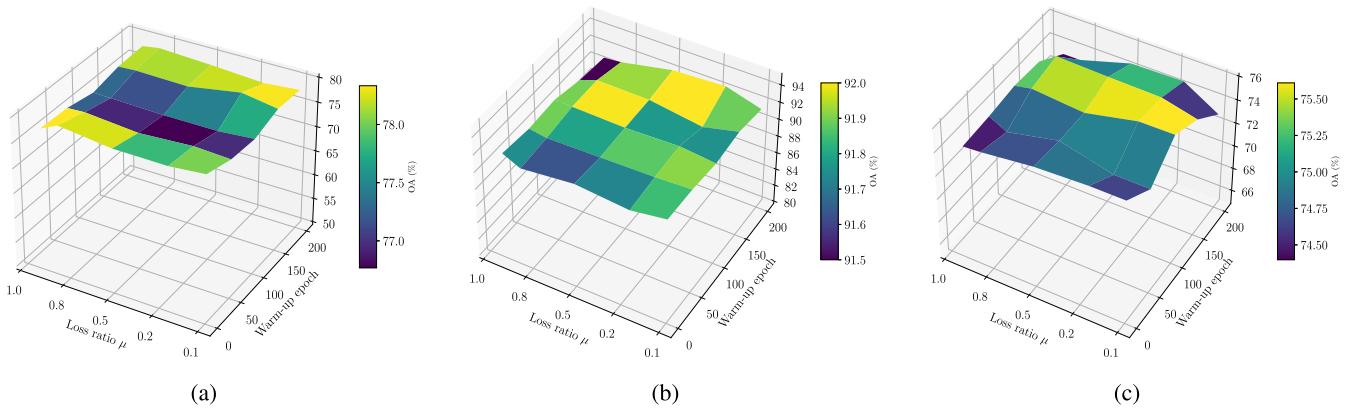


Fig. 9. Influence of μ and warm-up epoch on the performance of our method on three datasets: (a) IP, (b) SA, and (c) HC.

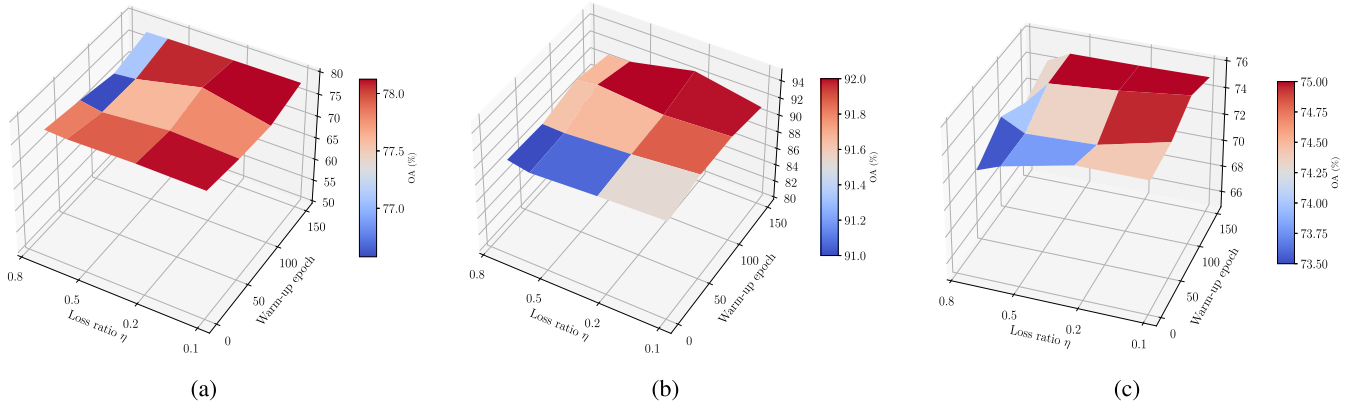


Fig. 10. Influence of η and warm-up epoch on the performance of our method on three datasets: (a) IP, (b) SA, and (c) HC.

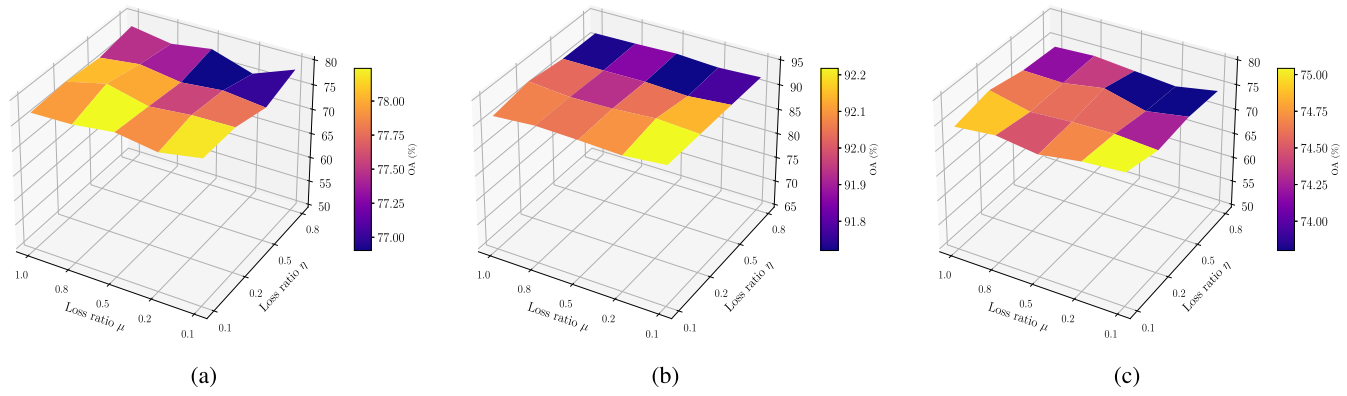


Fig. 11. Influence of μ and η on the performance of our method on three datasets: (a) IP, (b) SA, and (c) HC.

configurations, further validating its robustness to hyperparameter variations.

F. Analysis of Classification Performance With Different Numbers of Training Data

To evaluate the proposed method's robustness under limited labeled samples, we conduct experiments with 1–5 annotated samples per class in the target domain. Fig. 12 shows the OA and standard deviations of the results on the IP, SA, and HC datasets. It can be observed that the traditional deep learning method, SSRN, exhibits substantial instability, with large standard deviations, and its performance tends to decrease as the number of labeled samples increases. In contrast, the FSL methods, which employ a meta-learning strategy, show a consistent improvement in OA when more training data are used. Moreover, it is observed that our method outperforms others in most cases, demonstrating the superior performance of our method with limited training data.

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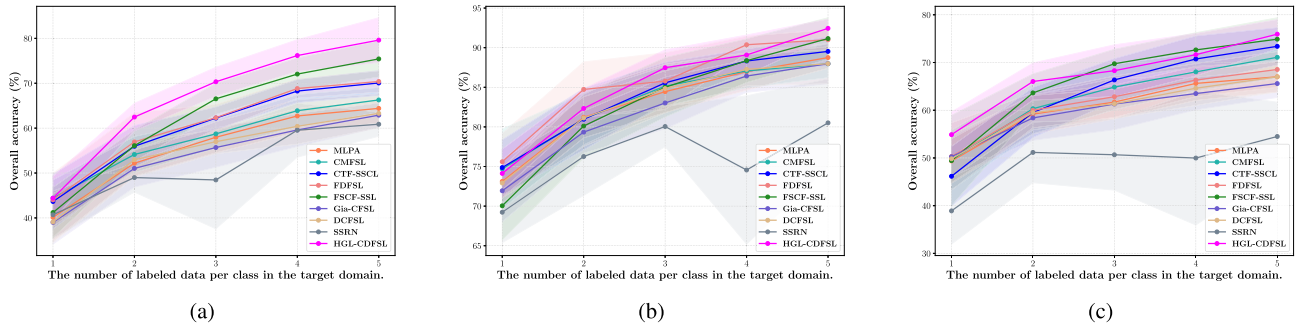


Fig. 12. OA (%) of various methods on the target datasets with different numbers of labeled samples: (a) IP, (b) SA, and (c) HC.

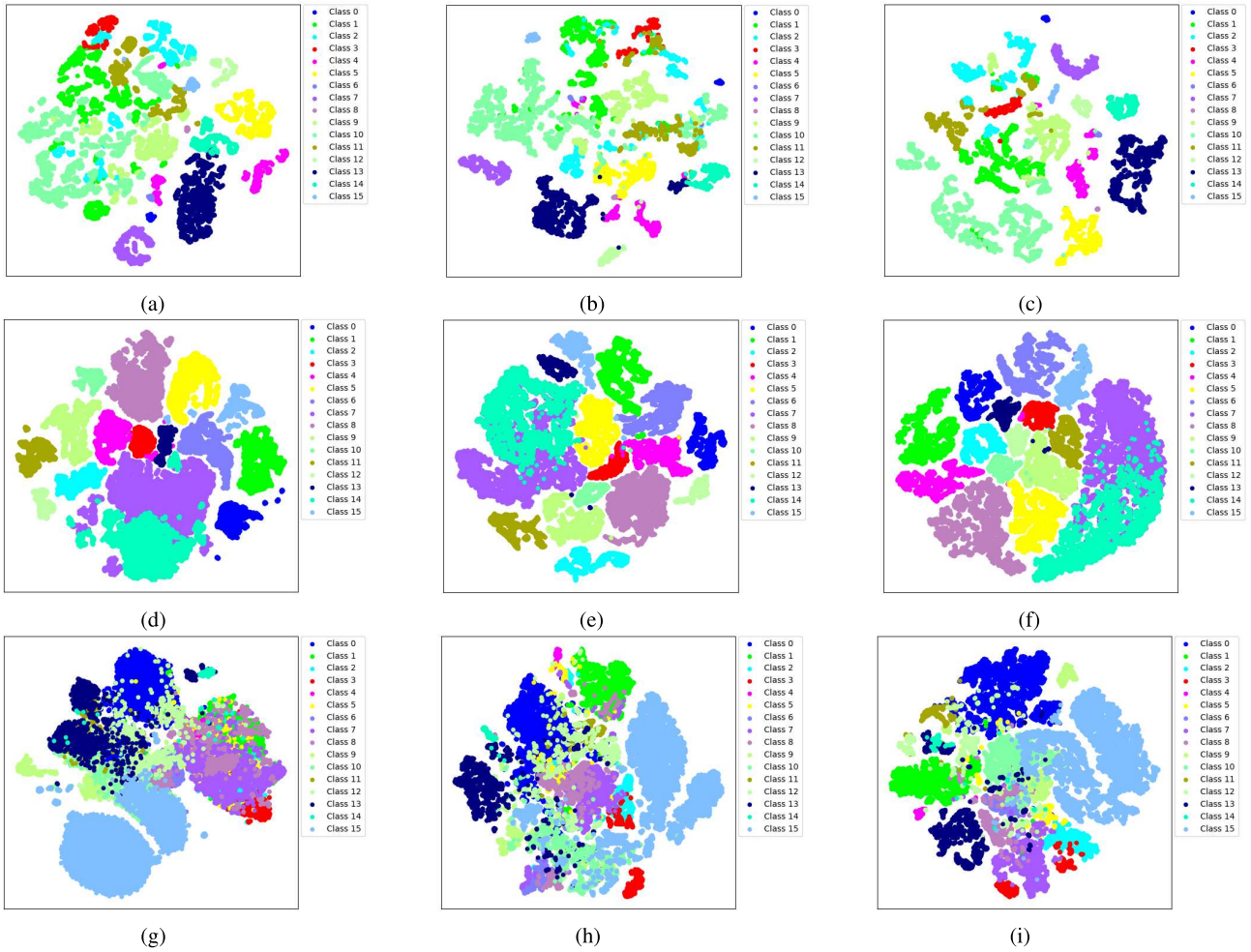


Fig. 13. Feature visualization via t-SNE on three datasets. (a) MLPA features on IP, (b) FSCF-SSL features on IP, (c) HGL-CDFSL features on IP, (d) MLPA features on SA, (e) FSCF-SSL features on SA, (f) HGL-CDFSL features on SA, (g) MLPA features on HC, (h) FSCF-SSL features on HC, and (i) HGL-CDFSL features on HC.

G. Visualization of the Features

We visualize the features extracted by different methods using t-distributed stochastic neighbor embedding (t-SNE) in Fig. 13, across three datasets. Two representative baselines, MLPA and FSCF-SSL, are selected for comparison. The results demonstrate that the features learned by our method exhibit better class separability than those of the other two methods, particularly on the IP and SA datasets. On the HC dataset, the distinction between the 8th, 12th, and 13th classes is also more evident in our method compared to MLPA and FSCF-SSL. These observations help explain the superior classification accuracy achieved by our method, as shown in Tables III–V.

H. Computational Complexity Analysis

We show the running time, the number of floating-point operations (FLOPs), and the number of parameters of different FSL methods in Table VIII. Three datasets are tested. As our

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TABLE VIII
COMPUTATION TIME (S), FLOPS (M), AND THE NUMBER OF PARAMETERS (M) OF DIFFERENT FSL METHODS ON THREE DATASETS

Dataset	Metric	Methods							
		DCFSL	Gia-CFSL	CMFSL	FSCF-SSL	FDFSL	CTF-SSCL	MLPA	HGL-CDFSL
IP	Training time	1160.93	5068.01	**735.85**	24177.80	1154.27	1005.48	1227.69	14205.18
	Testing time	9.69	1.56	3.28	5.56	1.72	**1.19**	1.74	5.11
	FLOPs	43.16	43.28	27.13	200.63	43.16	**6.35**	43.24	222.52
	#Params	**0.07**	1.08	0.61	3.30	0.09	0.40	**0.07**	3.34
SA	Training time	1301.42	2916.92	**891.59**	42001.34	1225.48	1300.89	1216.33	18835.27
	Testing time	4.72	8.64	**3.61**	24.98	8.95	17.04	7.80	28.54
	FLOPs	43.20	43.31	27.13	200.70	43.19	**6.35**	43.27	222.95
	#Params	**0.07**	1.08	0.61	3.34	0.09	0.41	**0.07**	3.34
HC	Training time	1023.29	3040.06	1175.37	69839.82	**641.26**	2436.57	1399.67	13613.84
	Testing time	5.71	3.71	4.21	79.14	7.51	**3.21**	6.83	18.44
	FLOPs	43.76	43.88	27.13	201.30	43.76	**6.95**	43.84	230.57
	#Params	**0.08**	1.09	0.61	3.42	0.09	0.41	**0.08**	3.35

TABLE IX
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON PAVIA UNIVERSITY

Methods	OA (%)	AA (%)	κ (%)
FSCF-SSL [36]	87.95±2.72	91.08±2.21	84.42±3.37
FDFSL [30]	82.78±2.61	85.20±2.78	77.75±3.15
CTF-SSCL [61]	82.60±3.71	85.05±1.65	77.54±4.43
MLPA [62]	81.22±3.34	81.95±2.31	75.71±3.84
HGL-CDFSL	**89.27±2.87**	**91.70±1.51**	**86.12±3.50**

spatial feature extractor at the first stage is only pretrained once on the natural RGB image dataset, which is reusable for different target datasets at the second stage, we mainly focus on the running time and the computational complexity of our model at the second stage. It is observed that FSCF-SSL and HGL-CDFSL run slower than other FSL methods, which can be mainly attributed to the large patch size that is required to adapt to the pretrained spatial feature extractor at the first stage, leading to a higher computational complexity. Compared with FSCL-SSL, our method is often faster during the training and testing stages. It should be noted that although the training time of our method is longer than that of other methods, the inference time remains comparable, while yielding significantly improved classification accuracy as shown in Tables III–V.

I. Generalization Capability

To further assess the generalization ability of the proposed method across different scenes, we conduct additional experiments on the Pavia University dataset, which contains diverse land-cover classes such as asphalt, meadows, gravel, trees, metal sheets, bare soil, bitumen, bricks, and shadows. We compare our method with four state-of-the-art approaches: FSCF-SSL, FDFSL, CTF-SSCL, and MLPA. The results are presented in Table IX. As shown, our method consistently achieves the highest performance in terms of OA, AA, and κ, demonstrating its robustness and effectiveness across diverse HSI domains.

V. CONCLUSION

In this article, we propose a novel CDFSL framework, HGL-CDFSL, for HSIC. To the best of our knowledge, this is the first method that simultaneously leverages both natural RGB images and hyperspectral data as heterogeneous source domains to learn transferable spatial–spectral meta-knowledge for guiding HSIC in the target domain. To mitigate domain shift, we design a DA strategy that operates from global to local levels. To address the severe spectral variability under limited target samples, we incorporate a self-supervised contrastive consistency strategy to promote intraclass compactness. Extensive experiments demonstrate that the proposed HGL-CDFSL consistently outperforms existing baselines under various few-shot settings.

While our method yields consistently improved classification accuracy, there remain opportunities to further improve its performance and efficiency. First, the relatively high model complexity may limit its deployment in resource-constrained scenarios. This motivates future integration with model compression or lightweight backbone techniques. Second, the current spatial–spectral network employs a relatively simple architecture, which may limit its ability to capture rich multiscale and long-range dependencies. To address this, advanced modules such as multiscale attention or transformer-based encoders can be incorporated to enhance spatial–spectral feature extraction and fusion.

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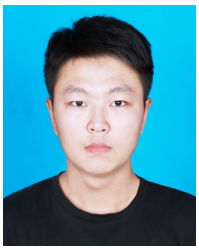
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Shaoguang Huang (Member, IEEE) received the M.S. degree in telecommunication and information systems from Shandong University, Jinan, China, in 2015, and the Ph.D. degree in computer science engineering from Ghent University, Ghent, Belgium, in 2019.

He is currently a Full Professor with the School of Computer Science, China University of Geosciences (Wuhan), Wuhan, China. His research interests include image processing, sparse representation, clustering, hyperspectral image analysis, and machine learning.



Kun Zhao (Graduate Student Member, IEEE) received the B.E. degree from the Qilu University of Technology, Jinan, China, in 2023. He is currently pursuing the M.E. degree in China University of Geosciences (Wuhan), Wuhan, China.

His research interests include hyperspectral image few-shot learning.



Hongyu Chen (Member, IEEE) received the B.S. degree in electronic engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2016 and 2023, respectively.

He is currently an Associate Professor with the School of Computer Science, China University of Geosciences (Wuhan), Wuhan. His major research interests include hyperspectral image restoration, high-resolution remote sensing imagery panoptic parsing, and computer vision.

Dr. Chen received the First Place Prize in the Pairwise Semantic Stereo Challenge from the 2019 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest.



Hongyan Zhang (Senior Member, IEEE) received the B.S. degree in geographic information systems and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2005 and 2010, respectively.

From 2010 to 2022, he was a Researcher at the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University. Currently, he is the Dean of the School of Computer Science, China University of Geosciences (Wuhan), Wuhan. He has authored/co-authored more than 100 research papers. His research interests include high-dimensional data intelligent processing and agricultural remote sensing.

Dr. Zhang was a recipient of the General Program for Young Scientists (Category A) from the National Natural Science Foundation of China. He scored 1st in the Data Fusion Contest of 2019 and 2021 organized by the IEEE Image Analysis and Data Fusion Technical Committee. He serves as an Associate Editor for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and *Photogrammetric Engineering and Remote Sensing and Computers and Geosciences*. He is a reviewer for more than 40 international academic journals, including *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, and IEEE TRANSACTIONS ON IMAGE PROCESSING.

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